6 A rigid metal coil with 60 turns measuring 12.0 cm by 8.5 cm is suspended from a sensitive newton-meter, as shown in Fig. 6.1. The metal coil is initially above a region of uniform magnetic field of flux density *B*.

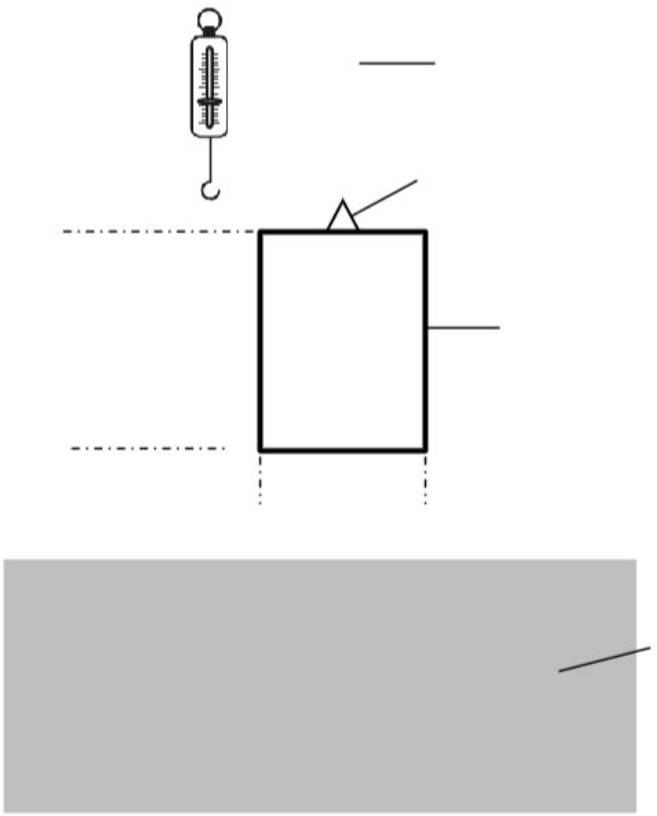


Fig. 6.1

(a) The coil is connected to a power supply so that there is a current of 6.0 A in the direction indicated in Fig. 6.1. The magnetic field is perpendicular to the page.

The coil is lowered into the uniform magnetic field until all of side PQ is just in the field. At equilibrium, the reading on the newton-meter is less than the reading when the coil is at the position shown in Fig. 6.1.

State and explain the direction of the magnetic field.

.....

.....

..... [2]

(b) When the side PQ of the coil is in the magnetic field in **(a)**, the newton-meter reading is 101.35 g.

The direction of the current is reversed and the current remains at 6.0 A. The newton-meter reading becomes 125.76 g at equilibrium.

Determine the magnetic flux density of the uniform magnetic field.

magnetic flux density = ... T [3]

(c) State and explain whether the reading on the newton-meter will differ from **(a)** if half of the coil is now in the magnetic field.

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..... [2]

[Total: 7]